Optoelectronics Lab – Experiment 3: Optical Signal Processing

Preparation – Task 1 and Questions 1–4 with worked answers

Name, Group: _____ Date: _____

The tasks and questions are quoted from the lab documentation (shaded boxes); the answers follow below each box. Simulations: LTspice 26.1.1 with the course models.

0. Background terms (requested in the lab documentation)

> **Note:** *Find out about the optical and electrical properties of photodiodes, phototransistors and luminescent diodes as well as the meaning of the terms "feedback", "non-linear system" and "time-invariant system".*

- **Feedback:** part of the output signal is returned to the input. With **negative** feedback it is subtracted, which stabilises the system and reduces the gain. With **positive** feedback it is added, which increases the gain and can make the system latch (bistability).
- **Non-linear system:** the superposition principle does not hold, i.e. the response to $a \cdot x_1 + b \cdot x_2$ is not $a \cdot y_1 + b \cdot y_2$. Diodes, saturation and switching are typical non-linearities.
- **Time-invariant system:** a time-shifted input produces the same, equally time-shifted output; the parameters do not change with time. A system with memory (hysteresis) can still be time-invariant.
- **LED:** forward-biased pn junction that emits light by recombination; P_{opt} roughly proportional to the current; forward voltage set by the band gap.
- **Photodiode:** reverse-biased pn junction; the photocurrent is proportional to the optical power ($I_{ph} = S \cdot P$), linear and fast (ns).
- **Phototransistor:** photodiode at the base–collector junction plus transistor gain B, so a much larger current but slower (µs) and less linear.

Task 1

Task 1a – LTspice simulation of the feedback optocoupler

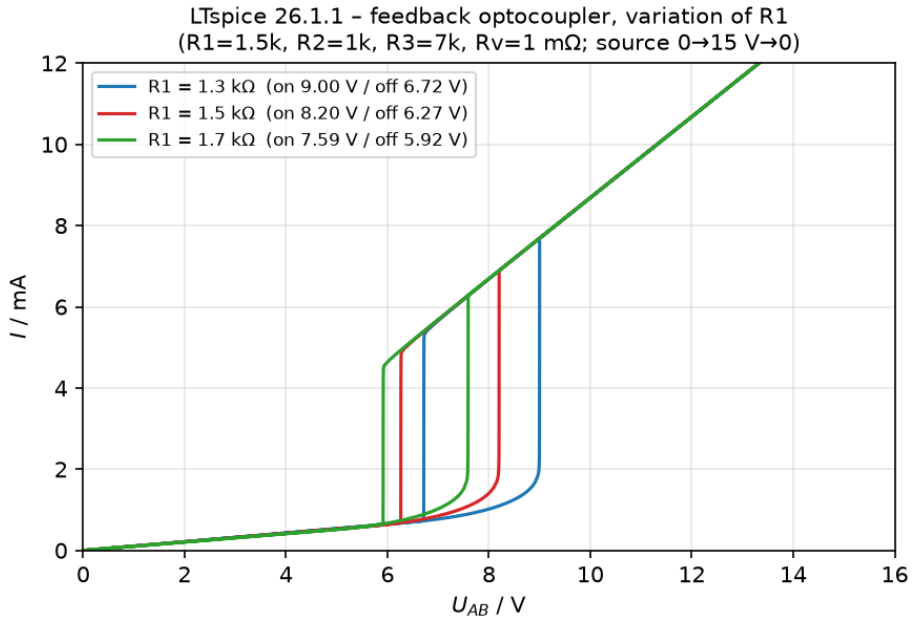
> **Task 1a)** *Carry out your own simulations with the LTspice model. Pay attention to the changes in the current-voltage characteristic curve when varying the resistors.*
> *– R_1: 1.5 kΩ, variation by ± 0.2 kΩ; – R_2: 1.0 kΩ, variation by ± 0.2 kΩ;*
> *– R_3: 7.0 kΩ, variation by ± 1.0 kΩ; – R_V: 1.0 mΩ, variation by + 1.0 kΩ*

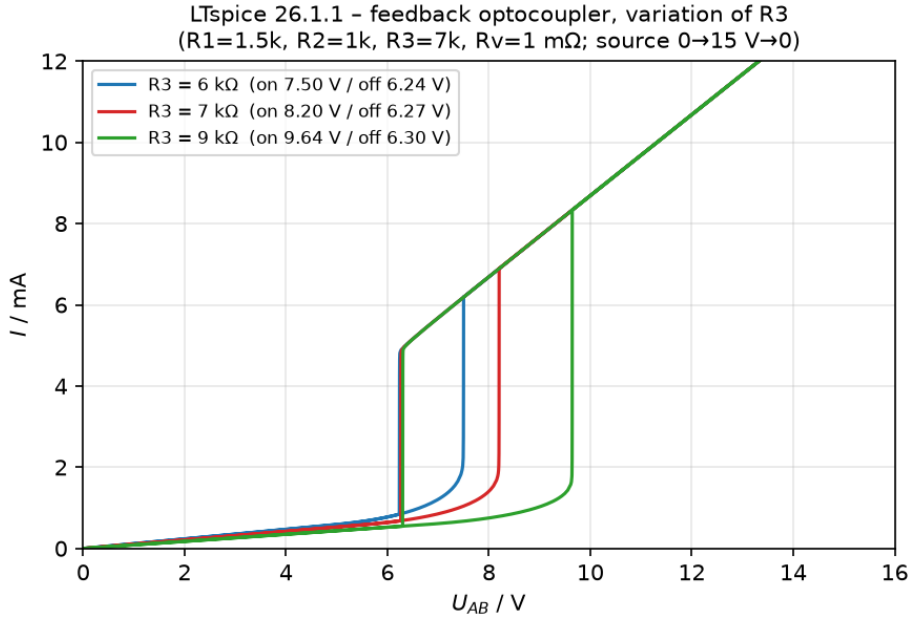
Simulation with the course files (Optokoppler_var_R1/R2/R3/Rv.asc); the source is swept 0 → 15 V → 0 (1 s per ramp) and I is plotted against U_{AB}.

Variation	Value	U_{on} / V (jump 1)	U_{off} / V (jump 2)	ΔU / V
R_1	1.3 kΩ	9.00	6.72	2.28
	1.5 kΩ	8.20	6.27	1.93
	1.7 kΩ	7.59	5.92	1.67
R_2	0.8 kΩ	7.83	5.30	2.53
	1.0 kΩ	8.20	6.27	1.93
	1.2 kΩ	8.60	7.23	1.37
R_3	6 kΩ	7.50	6.24	1.26
	7 kΩ	8.20	6.27	1.93
	9 kΩ	9.64	6.30	3.34
R_V (U_{source} at the jumps)	1 Ω	8.20	6.27	1.93
	100 Ω	8.40	6.75	1.65
	1 kΩ	–	–	no hysteresis

Table 1: Switching voltages of the feedback optocoupler (LTspice).

- **R_1:** the LED starts to conduct when $I \cdot R_1 \approx U_F$; on the OFF branch $I = U/(R_1+R_2+R_3)$, hence $U_{on} \approx U_F \cdot (R_1+R_2+R_3)/R_1$. Increasing R_1 lowers both switching voltages.
- **R_2:** in series in both states and dominant on the ON branch (slope $\approx 1/R_2$). Increasing R_2 raises both thresholds, flattens the ON branch and reduces the loop gain: the hysteresis narrows (2.53 V → 1.37 V).
- **R_3:** short-circuited by the phototransistor in the ON state, so it sets only the OFF-branch slope and U_{on}. U_{off} stays practically constant, so the hysteresis widens (1.26 V → 3.34 V).
- **R_V:** $U_{AB} = U_{source} - I \cdot R_V$ (load line). The jumps follow the load line with slope $-1/R_V$. The largest negative differential resistance is ≈ 0.82 kΩ; for R_V = 1 kΩ only one intersection exists and the complete S-curve (branch 3 of Fig. 5) is traced.


LTspice 26.1.1 – feedback optocoupler, variation of R1
(R1=1.5k, R2=1k, R3=7k, Rv=1 mΩ; source 0→15 V→0)
R1 = 1.3 kΩ (on 9.00 V / off 6.72 V)
R1 = 1.5 kΩ (on 8.20 V / off 6.27 V)
R1 = 1.7 kΩ (on 7.59 V / off 5.92 V)
I / mA
U_{AB} / V
0
2
4
6
8
10
12
14
16


LTspice 26.1.1 – feedback optocoupler, variation of R3
(R1=1.5k, R2=1k, R3=7k, Rv=1 mΩ; source 0→15 V→0)
R3 = 6 kΩ (on 7.50 V / off 6.24 V)
R3 = 7 kΩ (on 8.20 V / off 6.27 V)
R3 = 9 kΩ (on 9.64 V / off 6.30 V)
I / mA
U_{AB} / V
0
2
4
6
8
10
12
14
16

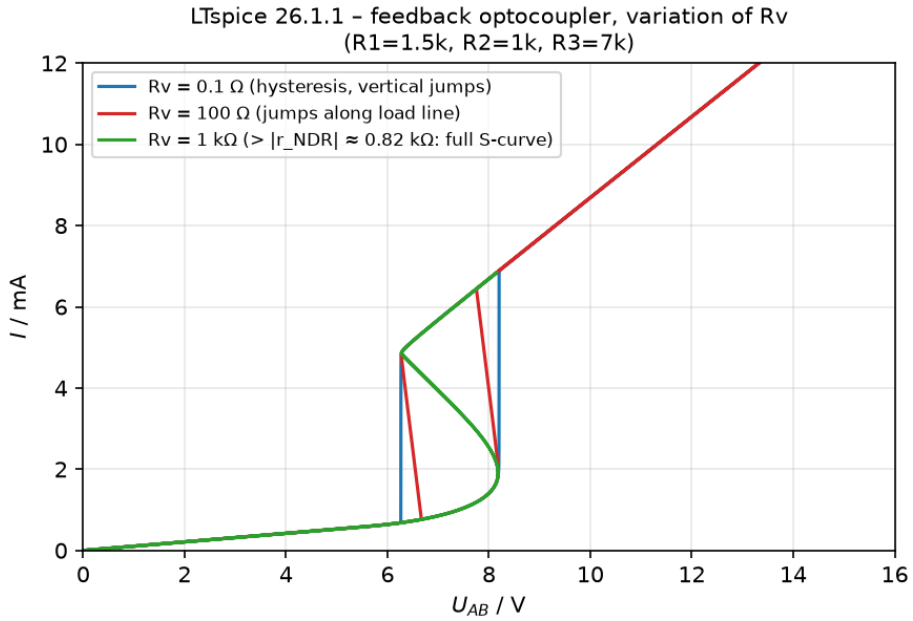


Fig. 1: I–U_{AB} characteristics simulated with LTspice (R_1, R_3 and R_V variations; R_2: see 05_Simulationsergebnisse).

Task 1b – Positive feedback

Task 1b) *Draw a block diagram showing positive feedback. Determine the gain factor A_{CL} of the closed circuit as a function of:*
– A_{OL} (open loop amplification factor) and – β (feedback factor).

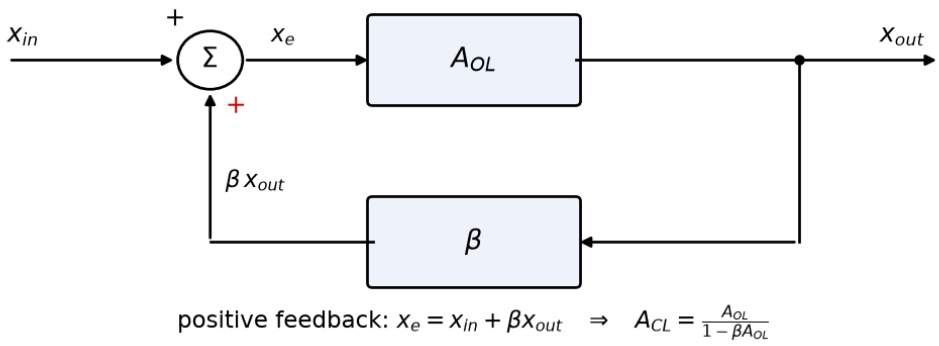


Fig. 2: Block diagram of positive feedback.

With $x_e = x_{in} + \beta \cdot x_{out}$ and $x_{out} = A_{OL} \cdot x_e$:

$$x_{out} = A_{OL} \cdot x_{in} + \beta \cdot A_{OL} \cdot x_{out} \quad \Rightarrow \quad \mathbf{A_{CL} = x_{out}/x_{in} = A_{OL} / (1 - \beta \cdot A_{OL})}$$

- $\beta \cdot A_{OL} < 1$: the gain is increased ($A_{CL} > A_{OL}$), the system is still stable.
- $\beta \cdot A_{OL} \to 1$: $A_{CL} \to \infty$; the system becomes unstable.
- $\beta \cdot A_{OL} > 1$: the output runs into saturation and stays there. This latching gives two stable states (bistability, hysteresis).

For comparison, negative feedback gives $A_{CL} = A_{OL}/(1 + \beta \cdot A_{OL})$. In the optocoupler, the loop LED → light → phototransistor → current through R_1 → LED is positive feedback. In the switching region $dU/dI \approx R_2 + R_3 \cdot (1 - CTR)$, which is negative for $CTR \cdot R_3 > R_2 + R_3$ (loop gain > 1). Because the CTR grows with the LED current, this condition is reached only above a certain current, and the circuit switches.

Task 1c – NPN transistor at the operating point

Task 1c) *Look at the circuit in Fig. 6. Draw the current-voltage characteristics (I_C via V_{CE}) for different base currents I_B. Then determine the characteristic curve for $R = R_{out} + R_E$ as a function of I_C and V_{CE}. Determine the formulas for calculating the open-circuit voltage $V_{CE,OC}$ and short-circuit current $I_{C,SC}$ as a function of V_B, R_E and R_{out}.*

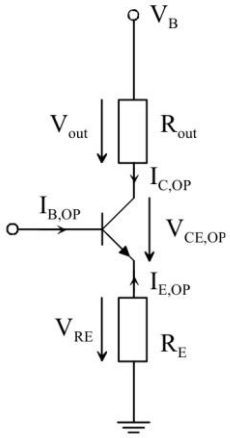


Fig. 6 of the lab documentation: NPN transistor at the operating point.

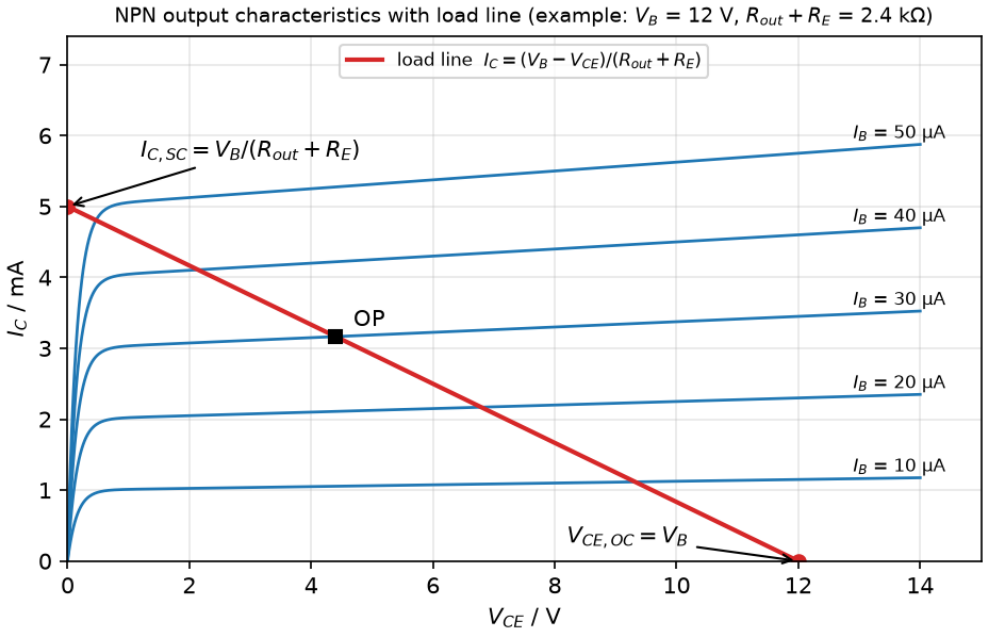


Fig. 3: Output characteristics $I_C(V_{CE})$ for different I_B with the load line of $R = R_{out} + R_E$.

Mesh equation: $V_B = I_C \cdot R_{out} + V_{CE} + I_E \cdot R_E$. With $I_E = I_C + I_B \approx I_C$ ($B \gg 1$):

$$\mathbf{I_C = (V_B - V_{CE}) / (R_{out} + R_E)} \quad \text{(straight line with slope } -1/R\text{)}$$

- Open-circuit voltage ($I_C = 0$): $\mathbf{V_{CE,OC} = V_B}$
- Short-circuit current ($V_{CE} = 0$): $\mathbf{I_{C,SC} = V_B / (R_{out} + R_E)}$

The operating point is the intersection of the load line with the characteristic for the given I_B. The same construction is used for the load line of R_V in the I–U characteristic of the feedback optocoupler.

Question 1

Question 1a) *What is the so-called forward voltage of an LED?*

Answer: The voltage across the LED at which it carries a noticeable current (usually specified at the nominal current, e.g. 20 mA). Because the characteristic is exponential, this voltage stays almost constant. It is set mainly by the band gap: $U_F \approx E_g/e \approx h \cdot c/(e \cdot \lambda)$. Typical values: IR (GaAs, 940 nm) ≈ 1.2 V, red ≈ 1.8–2.0 V, green ≈ 2.1–3 V, blue/white ≈ 3–3.5 V. Temperature coefficient ≈ −2 mV/K.

Question 1b) *What is the so-called current-transfer ratio in percent for an optocoupler with a phototransistor compared to one with a photodiode?*

Answer: $CTR = I_C/I_F \cdot 100\ \%$. With a phototransistor it is typically 20 % to several 100 % (PC817: 50–600 %, rank A 80–160 %), because the photocurrent is amplified by the transistor gain B. With a photodiode it is only about 0.1–1 %, because there is no internal gain. Trade-off: the phototransistor gives a high CTR but is slow (µs, Miller capacitance); the photodiode is fast (ns).

Question 2 (see LTspice simulation)

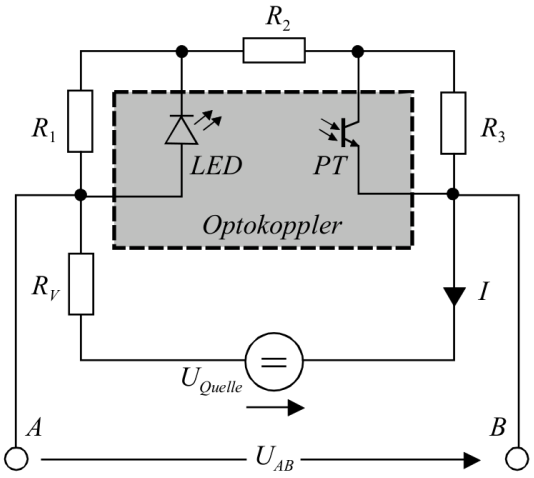


Fig. 4 of the lab documentation: feedback optocoupler.

Question 2a) *How does the characteristic curve of the feedback optocoupler come about?*

Answer: At low voltage the LED and the phototransistor are off and $I = U/(R_1+R_2+R_3)$ (flat branch). When $I \cdot R_1$ reaches U_F, the LED emits and the phototransistor starts to conduct. It shunts R_3, so the current rises, the LED becomes brighter and the transistor conducts more. This positive feedback makes the current jump to the ON branch. When the voltage is reduced, the phototransistor keeps R_3 short-circuited down to a lower voltage. The result is hysteresis, i.e. memory.

Question 2b) *Which components characterize the characteristic curve ranges (1) and (2)?*

Answer: Below jump (1) the curve is set by $R_1+R_2+R_3$ (LED and phototransistor off), and the jump is triggered by the LED threshold (U_F together with R_1). Above jump (2), on the ON branch, the curve is set by R_2, the LED and $U_{CE,sat}$. The jump down happens when the phototransistor can no longer stay saturated ($CTR \cdot I_{LED} < I - U_{CE}/R_3$), i.e. when the holding current is reached.

Question 2c) *What happens at the jump points?*

Answer: $dU/dI = 0$ (fold points) and the loop gain equals 1. The operating point merges with the unstable middle branch and disappears, so the circuit jumps along the load line to the other stable branch. In between lies a region of negative differential resistance, which is unstable under voltage control. Like a thyristor, the curve is S-shaped (current-controlled).

Question 2d) *What influence do the resistors R_1, R_2, R_3 and the series resistor R_V have on the curve behavior? Attention: the voltage U_{AB} is always measured between points (A) and (B)!*

Answer: See Task 1a. $R_1\uparrow$ lowers both thresholds; $R_2\uparrow$ raises them and flattens the ON branch; $R_3\uparrow$ raises U_{on} and widens the hysteresis; $R_V\uparrow$ tilts the load line, and for $R_V > |r_{NDR}| \approx 0.82\ k\Omega$ the S-curve becomes visible. Because U_{AB} is measured between A and B, R_V itself does not appear in the U_{AB} axis; it only decides which part of the curve can be reached.

Question 3

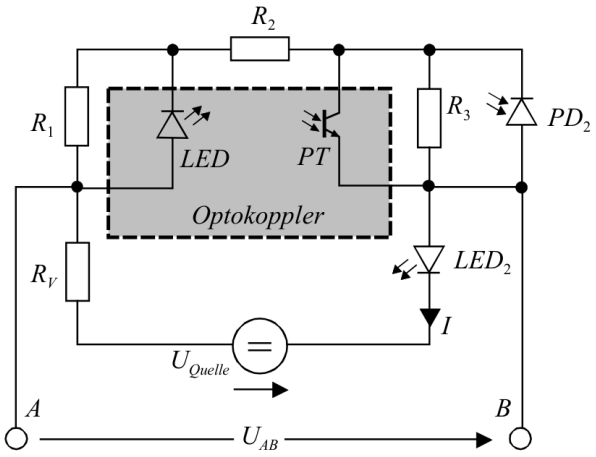


Fig. 7 of the lab documentation: extended circuit (PD_2 and LED_2 optically decoupled).

Question 3a) *What influence (see LTspice simulation) does the LED_2 have on the current-voltage characteristic?*

Answer: LED_2 is in series with the whole circuit. It shifts the characteristic to the right by its forward voltage (≈ 1.7–2 V; LTspice: U_{on} 8.20 → 9.93 V, U_{off} 6.27 → 8.05 V) and adds a small series resistance. It also serves as the optical output: it lights brightly only in the ON state.

Question 3b) *What is the relationship between current and optical power in a light-emitting diode and a photodiode?*

Answer: LED: $P_{opt} \approx \eta_{ext} \cdot (h \cdot \nu / e) \cdot I$, roughly linear (sublinear at high currents due to heating and droop). Photodiode: $I_{ph} = S \cdot P_{opt}$ with $S = \eta \cdot e/(h \cdot \nu)$ (≈ 0.5–0.6 A/W for Si near 850–900 nm), linear over many decades.

Question 3c) *How does illumination of PD_2 affect the feedback mechanism?*

Answer: The photocurrent flows in parallel to R_3 into the loop, so the total current and $I \cdot R_1$ rise. The LED switches on and the positive feedback latches the circuit in the ON state (SET); it stays ON after the light is removed. LTspice (with LED_2, U_{source} = 9 V, i.e. between the jump points 8.05 V and 9.93 V): an optical input equivalent to ≥ 0.3 mA input-LED current sets the memory (I: 0.96 → 5.9 mA); 0.2 mA is not sufficient.

Question 3d) *What is an LDR? What function does this component have when it is connected in parallel to R_1?*

Answer: LDR = light-dependent resistor (photoresistor, e.g. CdS): its resistance drops from MΩ in the dark to about 100 Ω under illumination. In parallel to R_1 it reduces the voltage across the LED, so the LED goes off, the phototransistor opens and the circuit returns to the OFF state (RESET). LTspice: reset for $R_{LDR} \lesssim 3\ k\Omega$ (5 kΩ is not sufficient).

Question 3e) *What is a Schmitt trigger and what can it be used for? What is the difference to a comparator?*

Answer: A comparator with positive feedback and two thresholds $U_{T+} > U_{T-}$ (hysteresis). It turns slow or noisy signals into clean digital edges and is used for debouncing, signal regeneration and relaxation oscillators. A plain comparator has only one threshold, so noise near the threshold causes multiple toggling (chatter). In the optical circuit, the memory and the Schmitt trigger differ only in the operating point: bistable in the dark gives a memory, only OFF in the dark gives a Schmitt trigger. LTspice (R_1 = 800 Ω, R_2 = 400 Ω, R_3 = 5 kΩ, U_{source} = 5 V, R_V = 0.1 Ω): ON at $I_{in} \approx 1.24$ mA, OFF at ≈ 1.01 mA; the hysteresis narrows with R_V and disappears for $R_V \gtrsim 300$ Ω.

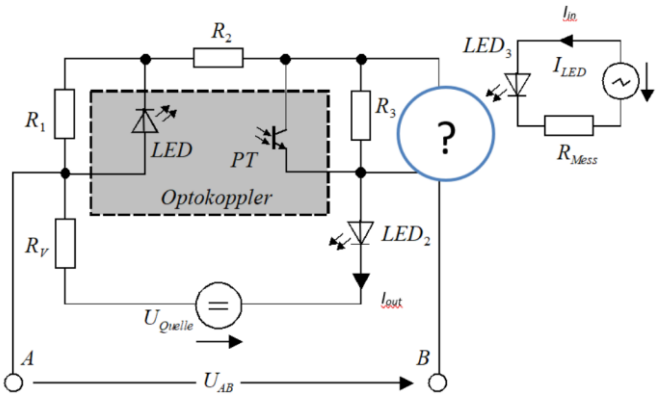


Fig. 10 of the lab documentation: optoelectronic Schmitt trigger.

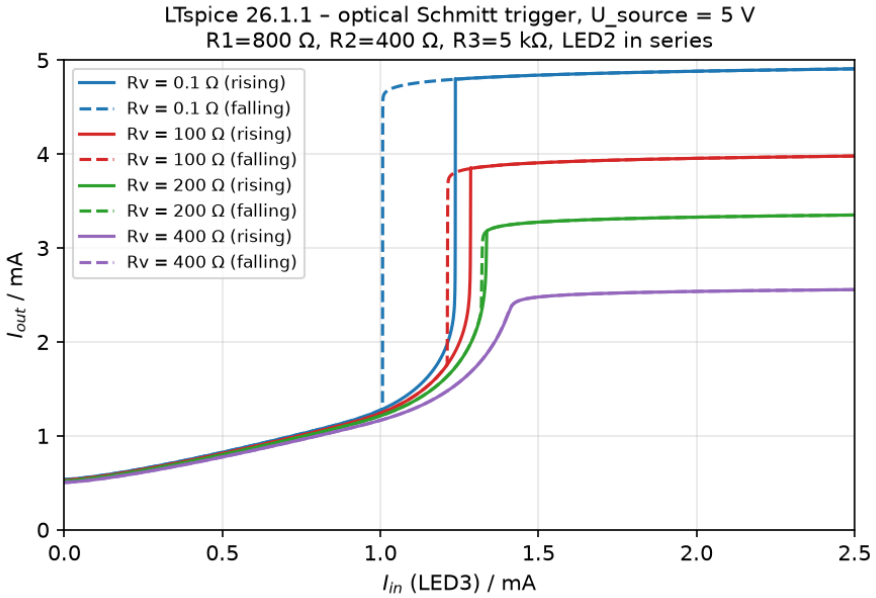


Fig. 4: Schmitt trigger simulated with LTspice.

Question 4

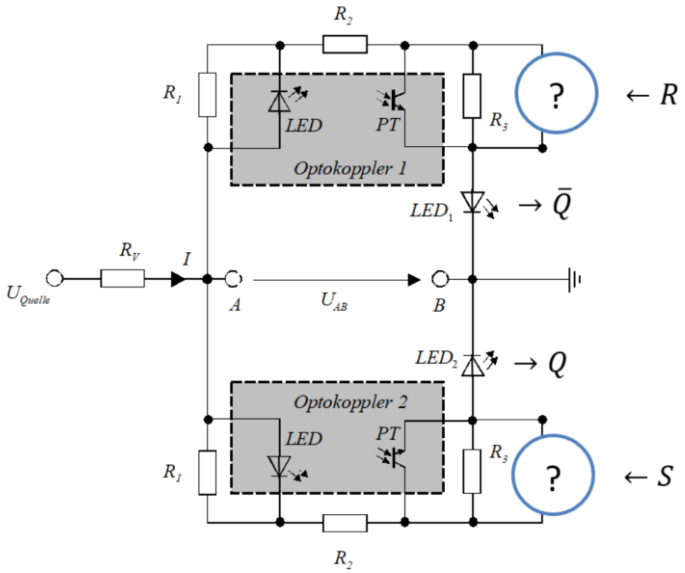


Fig. 8 of the lab documentation: two cells interconnected to an RS flip-flop.

Question 4a) *Which component is hidden behind the circled question mark?*

Answer: A photodiode (light-sensitive element) in parallel to R_3, illuminated by the R and S signals.

Question 4b) *What could the I-V characteristic curve of this network look like for $R_V = 0$?*

Answer: Both elements see the same voltage and switch independently. For identical elements the result is one hysteresis with double current (LTspice: both ON at 9.94 V, OFF at 8.05 V). For different elements it is a twin hysteresis with two steps (R_1 = 1k, R_4 = 2k: 8.64 V and 12.54 V). Both elements can be ON at the same time, so the network is no flip-flop.

Question 4c) *How must the load line for R_V be placed in the I-V characteristic curve in order to be able to switch only one switching element at a time?*

Answer: So that (i) the state with **one** element ON is stable, i.e. $U_{AB} = U_{source} - I_1 \cdot R_V$ lies between U_{off} and U_{on}; and (ii) the state with **both** ON is unstable, i.e. $U_{source} - 2 \cdot I \cdot R_V < U_{off}$. Switching on one element pulls U_{AB} so far down that the other cannot stay ON. LTspice ($R_1 = R_4 = 1.5$ kΩ, $R_2 = 1$ kΩ, $R_3 = 7$ kΩ, LEDs in series): with $R_V = 450$ Ω the network works as an RS flip-flop for $U_{source} \approx 10.6$–11.6 V (light ≙ 1 mA input-LED current; 0.3 mA is too weak). With $R_V = 0.1$ Ω both LEDs stay ON after S and R, so the flip-flop fails.

R	S	$\bar{Q}$ (LED_1)	Q (LED_2)
0	0	$\bar{Q}_n$ (hold)	Q_n (hold)
1	0	1	0
0	1	0	1
1	1	forbidden	forbidden

Table 2: Truth table of the RS flip-flop (theory). For R = S = 1 the optical circuit drives both LEDs ON ($Q = \bar{Q} = 1$); after both inputs are released the final state is a race (undefined).

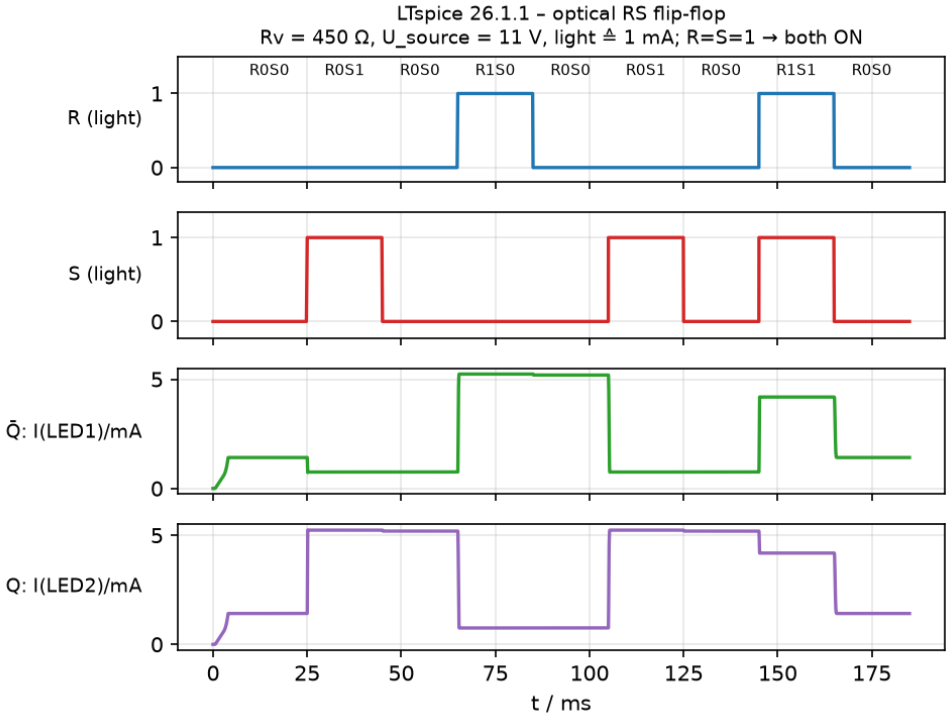


Fig. 5: Optical RS flip-flop simulated with LTspice (R_V = 450 Ω, U_{source} = 11 V).